Metlapalli Mehereswar

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Career Objective

Motivated college student with a strong interest in Web Development and Artificial Intelligence/Machine Learning. Proficient in Python and JavaScript with foundational knowledge of Data Structures and Algorithms. Seeking an entry-level Software Development Engineer (SDE) role or internship to apply technical skills to real-world projects and grow as a software engineer.

Education

Vellore Institute of Technology — Vellore, Tamil Nadu
Bachelor of Technology in CSE (AI/ML) — *2023–Present*
- CGPA: 8.89

FIITJEE Junior College — Hyderabad, Telangana
Higher Secondary — *2021–2023*

FIITJEE World School — Hyderabad, Telangana
Secondary — *2020*

Experience

Granules India Limited — May 2026 – Jul 2026
AI Intern — *Hyderabad, Telangana*
- Leveraged **Claude Code** and **Cursor** to accelerate feature implementation, debugging, refactoring, and documentation across AI and full-stack applications.
- Interacted with cross-functional departments including HR, IT, and EHS (Environment, Health & Safety) to contribute to full-stack feature development across the myGranules digital suite (8 enterprise platforms), including implementing secure **Role-Based Access Control (RBAC)** for the Project Management Office (PMO) platform, leveraging React, Vite, Node.js, and **Supabase**.
- Architected **Nyra conversational AI**, a secure internal assistant driven by self-hosted Qwen LLMs. Contributed to **model serving and inferencing** using SGLang on a 7-node GPU LAN.
- Engineered an in-house AI-led recruitment platform (Granules Recruit) featuring real-time AI interviews.
- Deployed the **Hermes** autonomous agent in Docker, connecting language models to live enterprise tools and data streams via custom Model Context Protocol (MCP) servers.
- Built autonomous AI-human workflows in **Paperclip**, engineering agents capable of planning and executing complex, multi-step tasks reliably in production.
- Spun up and managed cloud infrastructure from scratch on AWS EC2, automating deployments by configuring **CI/CD pipelines** via GitHub, Linux servers,and securely bridging cloud apps with on-premise GPU fleets.

Projects

Agentic SDLC Orchestrator | *Python, FastAPI, LangGraph, React, TypeScript, Vite, WebSockets, Gemini, Ollama*
- Architected an autonomous multi-agent software development platform that transforms a product brief into a PRD, system design, sprint plan, generated application code, and test suite through a 6-stage pipeline of 10 specialized LLM agents.
- Implemented self-correcting agent workflows with semantic validation, reviewer agents, and Human-in-the-Loop (HITL) approval gates supporting pausable and resumable executions using SQLAlchemy and Alembic.
- Built a provider-agnostic LLM layer supporting Gemini, Ollama, Groq, and OpenRouter with configurable per-stage routing, enabling hybrid local/cloud inference while enforcing structured outputs using Pydantic schemas.
- Developed a React + TypeScript dashboard with real-time pipeline visualization using Server-Sent Events and WebSockets.

Real-Time PPE Compliance Detection System | *Python, PyTorch, OpenCV, torchvision, MobileNetV2, Scikit-learn*
- Developed a real-time computer vision system to detect workers and verify compliance with mandatory Personal Protective Equipment (PPE), including helmets, safety vests, gloves, boots, and masks.
- Built an end-to-end inference pipeline incorporating image preprocessing, object detection, feature extraction, and PPE classification to identify compliant and non-compliant workers from live video streams.

- Optimized the model using transfer learning with MobileNetV2 and PyTorch, enabling efficient real-time inference while maintaining high classification performance.
- Designed the system for deployment in industrial environments to support automated workplace safety monitoring and compliance reporting.

StudyAI — AI-Powered PDF Study Assistant | *Python, FastAPI, Groq LLM, JavaScript, ReactJS*

- Built a full-stack web app that converts PDFs into study notes, flashcards, MCQ quizzes, and mind maps powered by Llama models via the Groq API. Engineered a real-time chat interface for document Q&A, highlight-to-explain feature, and interactive quiz scoring with instant feedback.

Smart Waste Segregation Using Machine Learning | *TensorFlow, Keras, MobileNetV2, OpenCV, Deep Learning*

- Developed an image-based waste classification system using transfer learning with MobileNetV2, incorporating dataset merging, preprocessing, normalization, and data augmentation to improve robustness. The model was fine-tuned using TensorFlow/Keras with the Adam optimizer, achieving approximately 92–95% validation accuracy, and designed for real-time integration with camera-enabled or IoT-based smart bins.

Technical Skills

Languages: Java, Python, C/C++, SQL, JavaScript, HTML/CSS

Frameworks & Technologies: React, Next.js, Node.js, FastAPI, TensorFlow, PyTorch, Vite

Cloud & DevOps: AWS (EC2),Git, Docker, Linux, Nginx, GitHub Actions, CI/CD

Databases: PostgreSQL, Supabase

AI/LLM: Qwen, Llama, SGLang, Model Context Protocol (MCP), Groq API

Developer Tools: GitHub, VS Code, Claude Code, Cursor, Figma, Wix Studio, Webflow

Libraries: NumPy, pandas, OpenCV, Scikit-learn, Matplotlib

Core CS: Data Structures and Algorithms, Database Management Systems, Object-Oriented Programming

Certifications

Machine Learning Specialization
DeepLearning.ai

Generative AI: Foundation Models and Platforms
IBM

Generative AI Leader Professional Certificate
Google